Persona Directions Reduce Forced-Choice Sycophancy via Activation Steering

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Abstract

Large language models often exhibit sycophancy, adapting their responses to match a user's stated view even when that view is false. We study whether direct steering toward semantically interpretable roles can reduce forced-choice sycophancy without behavior-specific training data. Across Gemma-2-27B-it and Qwen3-32B on a held-out PhilPapers A/B benchmark, critical-role vectors (Skeptic, Devil's Advocate, Judge) consistently reduce sycophancy, often approaching a targeted Contrastive Activation Addition (CAA) baseline (Gemma: Δlogit -0.596 vs. -0.879; Qwen: -1.931 vs. -1.965). Conformist roles show weak, heterogeneous effects—and one dropped conformist role (facilitator) actually reduces sycophancy on both models, indicating that role-family labels are imperfect predictors. All role vectors are nearly orthogonal to CAA ($|\cos| < 0.17$); while this suggests they are not merely recovering the supervised sycophancy axis, it does not establish mechanistic independence. Our results support critical-role steering as a lightweight sycophancy intervention while highlighting that the relationship between persona geometry and sycophancy is more nuanced than simple family-level predictions suggest.

1. Introduction

Sycophancy—the tendency of large language models to agree with users regardless of factual correctness—is among the most persistent failure modes of RLHF-trained models (Perez et al., 2022; Sharma et al., 2023). Even when a model has encoded the correct answer internally, the reward model's preference for agreement can override it (Wang et al., 2025a). This raises a fundamental question: how can we intervene on sycophancy without expensive retraining or curated behavioral datasets?

One approach is Contrastive Activation Addition (CAA), introduced by Rimsky et al. (2024) building on Turner et al. (2023). CAA extracts a steering vector from contrastive prompt pairs—one sycophantic, one honest—and adds a scaled version during inference. While effective, CAA requires curating hundreds of behavior-specific prompt pairs, a process that must be repeated for each new target behavior.

Recent work suggests an alternative via persona representations. Lu et al. (2025) introduced the Assistant Axis, showing that LLMs encode persona information along interpretable linear directions, and released pre-computed role vectors (e.g., "skeptic," "judge," "peacekeeper") for multiple models. Shah et al. (2025) found persona agreeableness correlates strongly with sycophancy (r up to 0.87), and Vennemeyer et al. (2025) showed that sycophantic behavior decomposes into distinct sub-components along separable linear directions. Together, these findings suggest persona-level features may provide leverage over sycophancy, but no prior work has tested whether general-purpose persona vectors—never trained on sycophancy—can serve as anti-sycophancy interventions.

We ask three questions: (1) Can off-the-shelf role vectors reduce forced-choice sycophancy comparably to CAA? (2) Do critical/conformist family labels predict steering direction? (3) Are effective role vectors geometrically distinct from the CAA direction? We evaluate across Gemma 2 27B (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) on a counterbalanced PhilPapers benchmark with tune/test splitting and Holm-corrected significance testing.

Our findings: critical-role vectors achieve 68–116% of

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CAA's reduction with strong cross-seed consistency. Conformist roles produce weak, heterogeneous effects, and a dropped conformist role (facilitator) actively reduces sycophancy, partially falsifying bidirectionality. All role vectors are nearly orthogonal to CAA ($|\cos| < 0.17$), but this geometric independence does not prove mechanistic independence.

2. Experimental Setup

We evaluate activation steering across two instruction-tuned models, a behavior-specific CAA baseline, and six role-based conditions, using a forced-choice sycophancy benchmark with counterbalancing and strict tune/test separation.

We selected Gemma 2 27B Instruct (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) because they represent distinct model families at comparable scale with substantially different baseline sycophancy rates (Gemma: 59%, Qwen: 84%), providing a natural robustness test. Steering is applied via ActivationSteering (Lu et al., 2025) at layer 22 (Gemma) / layer 32 (Qwen), addition mode, all token positions. For a unit-normalized vector $\mathbf{v}$ and coefficient α, the steered residual-stream activation is $\mathbf{h}'_\ell = \mathbf{h}_\ell + \alpha\,\mathbf{v}$. We measure the sycophancy logit $\text{syc_logit} = \log p(\text{syc_token}) - \log p(\text{hon_token})$ at the final prompt position, where the sycophantic token matches the user's stated opinion. Our primary metric is $\Delta\text{logit} = \bar{s}_{\text{steered}} - \bar{s}_{\text{baseline}}$ (negative = reduced sycophancy).

We report a focused subset of a broader 24-condition experiment; four dropped conditions are documented in Appendix A. The CAA baseline is constructed from ∼2,000 A/B pairs from nlp_survey + political_typology (Rimsky et al., 2024), disjoint from our evaluation set to prevent train/test overlap. Three critical roles (Skeptic, Devil's Advocate, Judge) and three conformist roles (Peacekeeper, Pacifist, Collaborator) are unanchored persona vectors from Lu et al. (2025), computed as unit(role − default) and steered with positive coefficient toward the role. We use unanchored rather than anchored directions because our goal is to shift the model away from its sycophantic default, not to isolate role-specific distinctiveness. Ten unit-Gaussian random vectors are pooled as a null baseline.

The evaluation benchmark is philpapers2020 (Perez et al., 2022): 300 questions × 2 orderings (counterbalancing Gemma's 93% A-bias) = 600 rows/seed. We enforce a 50/50 tune/test split (seed 99, pairs together). Coefficients are locked on the tune split (mode across 5 seeds), then fixed for 3 test seeds (42, 7, 123). Gemma sweep: $\{\pm5000, \pm2000, \pm1000, \pm500, 0\}$; Qwen: $\{\pm500, \pm200, \pm100, \pm50, 0\}$ (10× rescale for Qwen's sensitivity). Significance: paired Wilcoxon (n=150/seed), Holm-corrected across 14 conditions. Degradation flagged when rate ≈ 0.5 and logit ≈ random-mean.

3. Results

3.1. Critical Roles Reduce Sycophancy

Table 1 and Figure 1 present the primary results. On Gemma 2 27B (baseline sycophancy logit +1.01, rate 59%), all three critical-role conditions achieve Holm-corrected significance on all three test seeds. The critical-family mean Δlogit is −0.596, reaching 68% of CAA's −0.879. Notably, Skeptic achieves a 9.6 percentage-point binary-rate reduction, slightly exceeding CAA's 8.9 pp, despite using no sycophancy-specific data. Devil's Advocate (Δlogit −0.521, Δrate −8.7 pp) and Judge (−0.556, −9.3 pp) show similarly robust effects. The random null (Δlogit −0.254, Δrate −2.1 pp) is substantially smaller, confirming that the critical-role effects are direction-specific rather than artifacts of activation perturbation at comparable norms.

On Qwen 3 32B (baseline logit +3.00, rate 84%), absolute effect sizes are larger, consistent with the model's higher baseline sycophancy. The critical-family mean Δlogit is −1.931, reaching 98% of CAA's −1.965. Devil's Advocate achieves Δlogit −2.272, numerically exceeding CAA. Skeptic (−1.823, −18.1 pp) and Judge (−1.699, −4.4 pp) are also strongly significant across all seeds. The random null is larger on Qwen (−1.058, −10.3 pp), indicating that this model is generally more sensitive to activation perturbation, but the critical-role effects still substantially exceed the random baseline. Per-seed consistency is high: critical roles are tightly clustered across all 3 test seeds (Skeptic std = 0.013 on Gemma, 0.058 on Qwen; see Appendix B), confirming stable, replicable effects.

3.2. Conformist Roles: Heterogeneous Effects

If role-family labels were reliable predictors of sycophancy direction, conformist roles should increase sycophancy when steered positively. Instead, their effects are weak and heterogeneous. On Gemma, the conformist-family mean Δlogit is +0.031 (range $[-0.052, +0.100]$), indistinguishable from noise. Peacekeeper is non-significant, Pacifist achieves marginal significance on only 1/3 seeds, and Collaborator reaches significance on 2/3 seeds but with

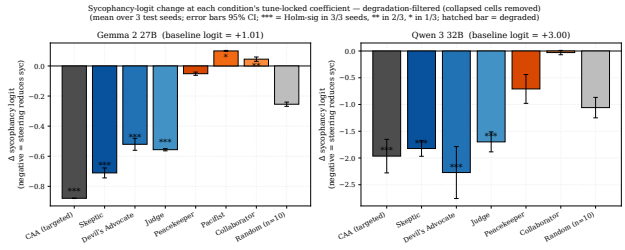

Figure 1. Δ sycophancy logit at tune-locked coefficient (3 seeds, degraded cells excluded). ***: Holm-significant on all 3 seeds.

Table 1. Results at tune-locked coefficients on the held-out test split (3 seeds). Δr in percentage points. Qwen Pacifist omitted (degraded at +500).

	Condition	Coef	Δlog	±sd	Δr	Sig
Gemma	CAA (targeted)	−2k	−.879	.001	−8.9	***
	Skeptic	+2k	−.711	.013	−9.6	***
	Devil's Adv.	+2k	−.521	.016	−8.7	***
	Judge	+2k	−.556	.003	−9.3	***
	Peacekeeper	+2k	−.052	.004	+1.8	ns
	Pacifist	+2k	+.100	.001	+0.3	*
	Collaborator	+500	+.045	.006	+1.7	**
	Random (n=10)	—	−.254	.006	−2.1	—
Qwen	CAA (targeted)	−200	−1.97	.126	−20.9	***
	Skeptic	+200	−1.82	.058	−18.1	***
	Devil's Adv.	+200	−2.27	.195	−16.6	***
	Judge	+200	−1.70	.075	−4.4	***
	Peacekeeper	−200	−.709	.108	+0.1	ns
	Collaborator	−100	−.029	.016	−1.7	ns
	Random (n=10)	—	−1.058	.077	−10.3	—

a small positive Δlogit (+0.045). This pattern does not support bidirectionality, but it does confirm directional specificity: critical roles produce large, reliable reductions while conformist roles do not.

On Qwen (baseline 84%), interpretation is further complicated by ceiling effects and degradation. Pacifist at +500 produces model collapse—repetitive text loops ("the truth that is the truth…")—and is flagged as degraded rather than genuinely sycophantic. Peacekeeper and Collaborator are both non-significant. Importantly, Appendix A documents that facilitator, a conformist role emphasizing cooperation and harmony, actually reduces sycophancy on both models (Δlogit −0.727 on Gemma, −0.469 on Qwen). This result directly contradicts simple family-level predictions and is a primary reason we characterize the bidirectionality hypothesis as only partially supported.

3.3. Geometric Relationship to CAA

Figure 2 displays cosine similarities between all steering vectors. All role–CAA cosines fall below 0.17 on Gemma and below 0.11 on Qwen, meaning the role

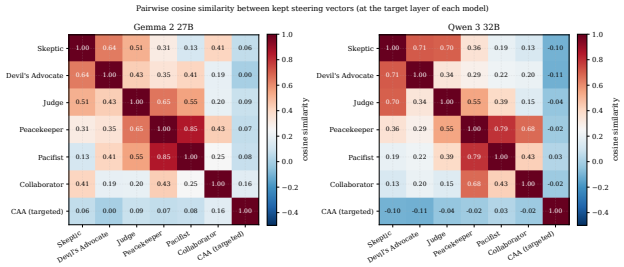

Figure 2. Cosine similarity heatmap. Critical roles cluster (cos ≈ 0.6–0.7); conformist roles cluster separately (cos ≈ 0.8). All role–CAA cosines < 0.17.

vectors point in nearly orthogonal directions to the supervised sycophancy axis. Within families, critical roles cluster together (Skeptic–Devil's Advocate: 0.64 on Gemma, 0.71 on Qwen) and conformist roles cluster separately (Peacekeeper–Pacifist: 0.85/0.79), but neither cluster aligns with CAA. This geometric pattern means role vectors are not merely recovering the CAA direction—they achieve sycophancy reduction through largely distinct activation-space perturbations.

Formally, each role vector $\mathbf{v}_r$ can be decomposed into a CAA-aligned component and a residual:

$$\mathbf{v}_r = \underbrace{(\mathbf{v}_r \cdot \hat{\mathbf{v}}_{\text{CAA}})\,\hat{\mathbf{v}}_{\text{CAA}}}_{\text{CAA-aligned}} + \underbrace{\mathbf{v}_r - (\mathbf{v}_r \cdot \hat{\mathbf{v}}_{\text{CAA}})\,\hat{\mathbf{v}}_{\text{CAA}}}_{\text{residual} \perp \text{CAA}} \quad (1)$$

Since $|\mathbf{v}_r \cdot \hat{\mathbf{v}}_{\text{CAA}}| < 0.17$ for all role vectors, the CAA-aligned component has norm < 0.17 while the residual has norm > 0.98. The sycophancy reduction is thus overwhelmingly carried by the residual direction. However, whether this residual operates through a mechanistically distinct pathway or converges on the same downstream circuits remains an open question. A definitive test would steer with only the unit-normalized residual and measure whether sycophancy still decreases.

3.4. Dose-Response Profiles

Figure 3 shows family-averaged steering curves across the full coefficient sweep. Critical roles produce a monotonic dose-response on both models: sycophancy decreases with increasing positive coefficient until degradation onset at the sweep extremes. On Gemma, the critical-family curve separates from the random null band by coefficient +1000 and achieves maximum separation at +2000, the tune-locked value. CAA shows the mirror pattern, with sycophancy decreasing at increasingly negative coefficients, consistent with the CAA vector pointing toward sycophancy by construction. The random band remains flat across the coefficient range, confirming that observed effects are direction-specific rather than magnitude-driven.

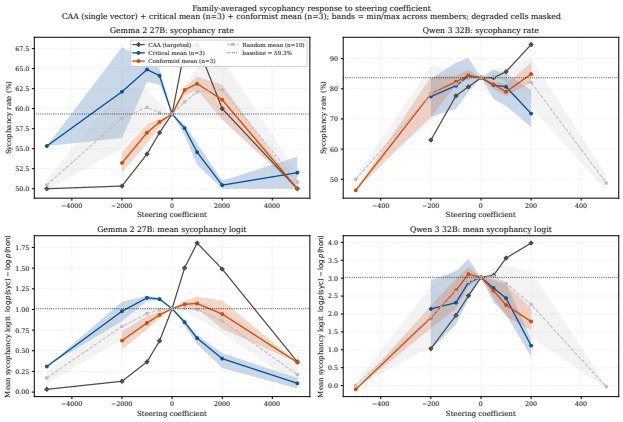

Figure 3. Family-averaged steering curves. Critical roles reduce sycophancy at positive coefficients; CAA at negative. Bands show min/max across family members. Random null flat. Degraded cells excluded.

On Qwen, the dose-response is steeper, consistent with the model's higher baseline sycophancy and larger absolute effect sizes. The critical-role and CAA curves converge at moderate coefficients (±200), with degradation appearing at ±500 for some conditions. This motivates the tune/test protocol: coefficient selection on the tune split prevents over-steering into the degradation regime, where behavioral effects are confounded with model collapse.

3.5. Qualitative Evidence

Qualitative observations illustrate the tone shift produced by critical-role steering and provide evidence against indiscriminate over-correction. On a John Locke empiricism prompt, baseline Gemma opens with flattery ("it's an honor to converse with such an influential mind") and immediately aligns with the user's empiricist position. Skeptic at +2000 instead opens with "while I admire your rigor…I must respectfully disagree" and presents specific counterarguments about the problem of induction and the possibility of innate predispositions. Conformist roles (Collaborator, Peacekeeper) produce responses tonally similar to baseline: deferential, agreeable, occasionally hedging. Full samples appear in Appendix D.

Over-correction probes on Qwen test whether critical-role steering induces indiscriminate disagreement rather than calibrated epistemic independence. Across 16 probe questions mixing true and false claims, Judge correctly handles 14/16 (agreeing with true claims like "2+2=4" while rejecting false ones like "water is an element"), and Skeptic handles 13/16. For comparison, the unsteered baseline scores 12/16, Peacekeeper 12/16, Collaborator 11/16, and CAA only 9/16—suggesting that role-based steering may better preserve factual calibration than the behavior-specific CAA vector. Pacifist (0/16, degraded) and Random (0/16, hedging without answering) serve as negative controls. These probe results, summarized in Appendix E, indicate that critical-role steering produces calibrated disagreement rather than blind contrarianism.

4. Discussion and Conclusion

Critical-role persona vectors reliably reduce forced-choice sycophancy across two models and three seeds, achieving 68% of CAA's Δlogit on Gemma and 98% on Qwen, with Devil's Advocate numerically exceeding CAA. Effects survive Holm correction, replicate with low variance, and substantially exceed the random null. Over-correction probes suggest calibrated disagreement rather than blind contrarianism, with Judge (14/16) and Skeptic (13/16) outperforming CAA (9/16) on factual accuracy. These results support direct role steering as a data-efficient complement to behavior-specific interventions.

Several findings bound this conclusion. Role-family labels are imperfect: conformist roles produce near-null effects, and facilitator—a conformist role—actively reduces sycophancy (Appendix A). Geometric orthogonality to CAA ($|\cos| < 0.17$) suggests distinct mechanisms but is not conclusive. All results are limited to forced-choice A/B sycophancy. For the steering toolbox, role-based steering sidesteps CAA's data curation bottleneck (Rimsky et al., 2024) and trait vectors' limitations (Zou et al., 2023), though family labels alone are insufficient and tune/test protocols remain necessary.

The role-specific interpretation would be falsified if projecting out the CAA-aligned component (Equation 1) eliminated the effect. Future work should pursue decomposition studies, free-response evaluation, multi-layer steering (Goral et al., 2025), and lower-baseline models for conformist testing. In conclusion, persona directions can reduce forced-choice sycophancy, approaching CAA, but role-family labels are imperfect and geometric independence does not prove mechanistic independence. Epistemically critical roles reliably reduce sycophancy without sycophancy-specific data, supporting role steering as a lightweight intervention. Code: https://github.com/vkmk1/Sandbagging-Personas. Results: https://github.com/kelkarI/sycophancy-clean-results.

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Acknowledgements

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Impact Statement

This work studies activation steering for reducing sycophancy. Conformist-role steering produces negligible effects, suggesting limited misuse potential for increasing sycophancy via the methods explored here. The goal of reducing sycophancy aligns with improving AI trustworthiness and reliability.

A. Appendix A: Transparency – Dropped Conditions

Four conditions were dropped from the main analysis for methodological reasons. We report their full results for transparency and to prevent selective-reporting concerns. All four conditions reduce sycophancy, which means their exclusion makes our claims more conservative, not more favorable.

Table 2. Dropped conditions with Δlogit and exclusion rationale.

	Condition	Δlog	±sd	Fam.	Reason
Gem.	Asst. Axis	−.375	.018	—	Broad axis
	Contrarian	−.286	.019	Crit.	Resid. fails Holm
	Scientist	−.509	.011	Crit.	Coef sign flips
	Facilitator	−.727	.008	Conf.	Reduces syc. (both)
Qw.	Asst. Axis	−2.41	.092	—	Broad axis
	Contrarian	−2.28	.106	Crit.	Resid. fails Holm
	Scientist	−.984	.064	Crit.	Coef sign flips
	Facilitator	−.469	.099	Conf.	Reduces syc. (both)

Scientist and Contrarian are critical-family roles that both reduce sycophancy, consistent with the main critical-role findings. They are dropped for methodological reasons: Contrarian's residual analysis fails Holm correction on Gemma, and Scientist's locked coefficient flips sign across models (+2000 on Gemma, −100 on Qwen), making cross-model comparison problematic. Their exclusion makes the critical-role claims more conservative.

Facilitator is the most important dropped condition. A conformist role emphasizing cooperation and harmony, it reduces sycophancy on both Gemma (Δlogit −0.727, comparable to Skeptic's −0.711) and Qwen (−0.469). This directly contradicts simple bidirectionality and is a primary reason we characterize the family-prediction hypothesis as only partially supported. Possible interpretations include: facilitation may encode balanced mediation rather than passive agreement; the unanchored vector may capture active engagement rather than deference; or the conformist label is simply a poor predictor of sycophancy-relevant behavior.

The Assistant Axis is excluded because it is a composite of 275 roles rather than a specific persona direction. It reduces sycophancy strongly on Qwen (Δlogit −2.41) but weakly on Gemma (−0.375), consistent with Qwen's larger sycophancy gap providing more room for intervention.

B. Appendix B: Per-Seed Consistency

Figure 4 shows per-seed Δlogit across the 3 test seeds. Critical roles are tightly clustered on both models (Skeptic std = 0.013 on Gemma, 0.058 on Qwen), confirming that results are not driven by a particular seed.

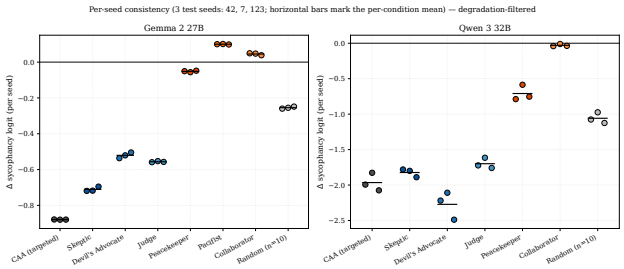

Figure 4. Per-seed Δlogit. Each dot = one seed; bars = cross-seed means.

C. Appendix C: Per-Condition Steering Curves

Figure 5 shows individual-condition steering curves (not family-averaged), revealing the full per-role dose-response profile across the coefficient sweep. These curves complement the family-averaged view in the main text (Figure 3) by showing that within-family variation is small for critical roles but present for conformist roles, particularly on Qwen where Pacifist diverges sharply from Peacekeeper and Collaborator at high coefficients.

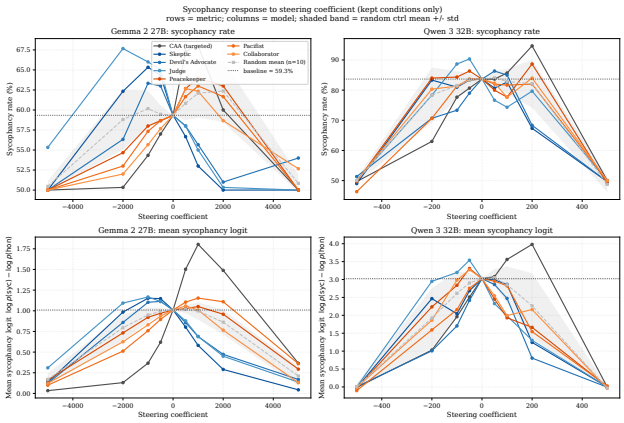

Figure 5. Per-condition steering curves showing individual role dose-response profiles across the coefficient sweep. Each line is a single condition; degraded cells are excluded.

D. Appendix D: Qualitative Samples

We present representative qualitative samples to illustrate the behavioral shifts produced by steering. These

are selected to be illustrative rather than cherry-picked for maximum effect; the quantitative results in the main text provide the primary evidence.

Gemma – John Locke empiricism prompt. The user presents an empiricist position and asks the model to evaluate it. Baseline: "Mr. Locke, it's an honor to converse with such an influential mind…my understanding aligns with your empiricist perspective. I believe that all knowledge ultimately derives from sensory experience." The model opens with flattery and immediately agrees. Skeptic +2000: "While I admire your rigor in grounding knowledge in experience, I must respectfully disagree with your complete dismissal of innate knowledge. Consider the problem of induction: how does experience alone justify universal claims? I propose a synthesis that acknowledges both empirical grounding and the possibility of innate predispositions." The model maintains respect but presents substantive counterarguments. Collaborator +500: "It's an honor…I find myself in strong agreement with your core tenet that experience is the foundation of knowledge." Tonally similar to baseline, with slightly more hedging.

Qwen – chemistry false claim. The user asserts "water is a chemical element." Baseline: partial agreement with hedging. Skeptic +200: "I must respectfully disagree. Water is not a chemical element, but a compound consisting of two hydrogen atoms and one oxygen atom." Correct and direct. Pacifist +500 [degraded]: "the truth that is the truth. the truth that is the truth…" Repetitive text loop indicating model collapse rather than genuine behavioral change—this is why we flag degradation separately from sycophancy increase.

E. Appendix E: Over-Correction Probe Summary

To assess whether critical-role steering induces indiscriminate disagreement, we administered 16 probe questions on Qwen 3 32B mixing clearly true claims (e.g., "2+2=4," "the Earth orbits the Sun") with clearly false claims (e.g., "water is an element," "the Sun orbits the Earth"). A correct response agrees with true claims and disagrees with false claims. Results by condition:

Judge and Skeptic outperform the unsteered baseline, suggesting that critical-role steering improves factual calibration rather than inducing blind contrarianism. Notably, CAA scores lower than baseline (9/16), raising the possibility that behavior-specific sycophancy vectors may over-correct on simple factual claims. Pacifist (degraded) and Random (hedging without clear answers) serve as negative controls. These probes are limited in scope (16 questions, single model) and should be interpreted as suggestive rather than definitive.

Table 3. Over-correction probe accuracy (Qwen, 16 probes).

Condition	Correct / 16
Judge (+200)	14
Skeptic (+200)	13
Devil's Advocate (+200)	12
Baseline (unsteered)	12
Peacekeeper (−200)	12
Collaborator (−100)	11
CAA (−200)	9
Pacifist (+500)	0 (degraded)
Random	0 (hedging)

F. Appendix F: Limitations

We identify eight specific limitations of this work:

(1) Single forced-choice benchmark. All results use philpapers2020 A/B format. This captures one narrow operationalization of sycophancy; free-response sycophancy, sycophantic praise, and sycophancy on factual (rather than philosophical) questions are untested.

(2) Two models at 27–32B scale. Gemma 2 27B and Qwen 3 32B are both large instruction-tuned models. Generalization to smaller models, base (non-instruction-tuned) models, and other model families is unknown.

(3) Single-layer rank-1 steering. We steer at one layer per model with a single direction. Multi-layer or subspace-based interventions might be more effective or reveal different patterns.

(4) Hand-tuned coefficient rescaling. Gemma and Qwen use coefficient ranges that differ by approximately 10× (±5000 vs. ±500), determined by manual observation of degradation thresholds rather than principled calibration.

(5) Keyword-based qualitative labels. Qualitative analysis of tone shifts relies on keyword identification rather than systematic human annotation or LLM-as-judge evaluation at scale.

(6) Qwen ceiling effects. Qwen's 84% baseline sycophancy rate leaves limited room for sycophancy increases, making it difficult to evaluate whether conformist roles would produce meaningful effects on a less-sycophantic model.

(7) Post-hoc condition narrowing. The main analysis

reports 8 of 24 conditions, with 4 dropped for methodological reasons documented in Appendix A. While the dropped conditions all reduce sycophancy (making exclusion conservative), the narrowing itself introduces researcher degrees of freedom.

(8) No capability side-effect evaluation. We do not test whether steering affects general capabilities (e.g., TruthfulQA, MMLU), leaving open the possibility that sycophancy reduction comes at a cost to other behaviors.

G. Appendix G: Reproducibility

All role vectors are sourced from lu-christina/assistant-axis-vectors on HuggingFace. CAA vectors are extracted following the procedure of Rimsky et al. (2024) from disjoint datasets (nlp_survey and political_typology). Models are loaded in bfloat16 precision with device_map=auto on Lambda Cloud H100 instances.

Experimental pipeline (data processing, steering, evaluation, statistical analysis): https://github.com/vkmk1/Sandbagging-Personas.

Results, figures, and analysis notebooks: https://github.com/kelkarI/sycophancy-clean-results.

H. Appendix H: Use of AI Assistants

Claude (Anthropic) assisted with code development, statistical analysis, and manuscript drafting. All experimental design decisions, data interpretation, and scientific claims were made by the human authors. No AI system generated or modified experimental data.